

USER STORIES DOCUMENT

File Compression Tool using Huffman Coding

User Stories

ID	Priority	User Story	Acceptance Criteria
US-01	High	As a user, I want to select a file from my computer so that I can compress it.	User can select a valid file; file name and size are displayed; invalid/unavailable files are rejected.
US-02	High	As a user, I want to compress a file using Huffman coding so that I can reduce its storage size.	System calculates frequencies, builds the Huffman tree, generates codes, encodes the data, and creates a compressed file.
US-03	High	As a user, I want the system to generate and display Huffman codes so that I can understand how the compression is performed.	Frequency data is calculated correctly; Huffman tree is generated; a binary code is assigned to each relevant byte/character; codes can be viewed.
US-04	High	As a user, I want to decompress a compressed file so that I can recover my original file.	User can select a valid compressed file; Huffman information is read/reconstructed; data is decoded; the original file is recreated.
US-05	High	As a user, I want compression to be lossless so that no information is lost from my original file.	Decompressed output matches the original file exactly; file contents remain unchanged; integrity verification succeeds.
US-06	High	As a user, I want to view compression statistics so that I can understand how effective the compression was.	Original size, compressed size, compression ratio, and percentage of space saved are displayed; compression time may also be shown.
US-07	Medium	As a user, I want to visualize the Huffman tree so that I can understand the relationship between frequencies and generated codes.	Generated tree is displayed with relevant byte/character and frequency information and corresponds to the generated codes.
US-08	Medium	As a user, I want to choose the output location so that I can save the compressed or decompressed file where I want.	User can select a destination; output is saved there; overwriting an existing file requires confirmation.
US-09	Medium	As a user, I want to receive meaningful error	Invalid input, corrupted files, and failed operations

		messages when an operation fails so that I know what went wrong.	produce appropriate messages without silently creating invalid output.
US-10	Low	As a user, I want to compress multiple files together so that I can reduce the effort of processing files individually.	User can select multiple valid files; all are processed successfully; file data is preserved.